

Machine Learning Engineer with 4+ years experience architecting scalable computer-vision and generative-AI solutions. Skilled in Python, PyTorch, FastAPI, and AWS services.

SKILLS

- **Programming:** Python (Expert), C++, SQL, Bash/Shell
- **ML & AI:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CatBoost, Transformers, Generative AI, NLP, Hugging Face
- **Computer Vision:** OpenCV, Pillow, scikit-image, MoviePy, YOLO, GroundingDINO, SAM, Detectron2
- **Web & APIs:** FastAPI, Django, Flask, Streamlit, REST APIs
- **Data & Databases:** NumPy, Pandas, PostgreSQL, MySQL, MongoDB, Excel
- **Cloud:** AWS (S3, EC2, Lambda, SQS, SageMaker), Azure, boto3
- **Tools & DevOps:** Git, Docker, CI/CD, MLflow, Airflow, Prefect, LangChain, Selenium, BeautifulSoup, Tableau, Power BI

PROFESSIONAL EXPERIENCE

ODN Digital Services Senior AI/ML Engineer Oct 2025 – Present

Agentic AI Platform for Automated Tool Orchestration

- Architecting an agentic AI system that enables consumers to interact with internal tools using natural language instructions.
 - Designed a multi-agent workflow: information-gathering agent → validation agent → tool-specific request generator → queue dispatcher.
 - Built an LLM-driven intent processor to map user requests to the appropriate internal tool (image processing, analytics, content generation, etc.).
 - Implemented an autonomous questioning module: the agent asks for and validates all mandatory parameters required by each tool (workflow details, consumer needs, execution constraints) before proceeding.
 - Engineered a generation layer that converts validated requirements into structured work-request payloads (not programming code) that follow the internal tool contract.
 - Integrated an automated queuing system to submit these generated work-request payloads into the task queue without human intervention, enabling end-to-end automation.
 - Applied Pydantic-based validation to ensure each payload is complete and tool-compatible, reducing execution failures and increasing reliability.
 - Improved non-technical user enablement by allowing users to trigger complex tool workflows using plain-language prompts.
- Technologies Used: Python, GPT-OSS, LangChain, FastAPI, Pydantic, AWS SQS, Internal Tool APIs

ODN Digital Services Junior Data Scientist May 2024 – Sep 2025

Automated Garment Attribute Extraction

- Leveraged the MMFashion cloth segmentation model to isolate garment regions, then refined masks with OpenCV and extracted cropped images.
 - Fed those crops into a GPT-4o LLM (via LangChain) to generate structured attributes (color, pattern, silhouette).
 - Modeled outputs with a Pydantic BaseModel schema and Pydantic validators, automatically flagging/logging malformed values and reducing manual annotation by 80%.
 - Deployed the FastAPI service on AWS EC2 to provide a scalable, production-ready API endpoint.
- Technologies Used: Python, MMFashion, OpenCV, LangChain, Pydantic, GPT-4o, FastAPI, AWS EC2

AI Content Generation Suite

- Architected and delivered a unified AI Content Generation Suite that automates both static infographic and dynamic video creation workflows for grocery and e-commerce catalogs.
 - **Grocery Infographics Module:**
 - Streamlit front-end accepts images and calls Claude 3.5 Sonnet to extract key benefits, product name, nutritional facts, and usage tips.
 - Dynamically composes high-fidelity infographic templates using Pillow, overlaying text, icons and brand colors.
 - Validates and normalizes all extracted fields via Pydantic schemas, logging anomalies for review and reducing manual infographic creation time by 90%.
 - **E-commerce Video Generator Module:**
 - Django REST APIs ingest FSN lists from Excel uploads, perform web scraping for product details, then use Claude 3.5 Sonnet to script feature-rich voice-over and on-screen captions.
 - Orchestrates MoviePy + ffmpeg pipelines to assemble 15–30 sec promotional videos with randomized transition and animation templates, delivering 50+ unique videos per hour.
 - Incorporates Pydantic for payload validation and error handling, achieving a 90% reduction in manual editing efforts.
 - Hosted both Streamlit and Django services on AWS EC2 instances to ensure reliable, scalable production deployment.
- Technologies Used: Python, Streamlit, Django REST Framework, Claude 3.5 Sonnet, Pydantic, Pillow, MoviePy, ffmpeg, AWS EC2

Intelligent Image Processing Pipeline

- Leveraged Florence and SAM2 to detect and segment barcodes, then decoded SKUs with PyBarcode after OpenCV-based preprocessing.
- Applied cosine similarity on image embeddings to identify and remove duplicate images, ensuring data cleanliness.
- Extracted human keypoints via MediaPipe and trained a PyTorch CNN on these features for accurate pose estimation.

- Employed OpenCV for grid-based segmentation on person and product images, programmatically renaming files based on grid coordinates, detected pose, and SKU metadata to streamline dataset organization.
- Achieved a 75% reduction in end-to-end processing time compared to manual workflows.
- Deployed the pipeline on an on-premises server and integrated AWS SQS for scalable task queuing.
Technologies Used: Python, Florence, SAM2, OpenCV, PyBarcode, MediaPipe, PyTorch, scikit-learn, AWS SQS

Optimized Background Removal

- Leveraged Florence and SAM2 to segment face and hair regions, generating high-fidelity masks for complex hairstyles and fine facial features.
- Fine-tuned BirefNet on a custom 10,000-image portrait dataset, boosting hair-mask IoU from 85% to 95% and reducing halo artifacts by 40%.
- Applied multi-scale context analysis and alpha matting, using OpenCV morphological filters and gradient-based refinement to smooth mask boundaries.
- Integrated adaptive post-processing—including color correction and bilateral filtering—to seamlessly blend subjects onto new backgrounds.
- Exposed the pipeline as a FastAPI microservice with Uvicorn, enabling concurrent processing of 20 images/sec in production.
- Achieved a 60% reduction in manual touch-up time and 30% faster end-to-end background removal compared to the baseline.
Technologies Used: Python, PyTorch, BirefNet, Florence, SAM2, OpenCV, FastAPI, Uvicorn, AWS EC2

WhatsApp Customer Support Chatbot

- Built an LLM-driven WhatsApp chatbot to handle client queries, order tracking, and service requests, reducing manual intervention by the customer support team.
- Implemented an agentic 4-stage workflow: intent classification, ambiguity detection, clarifying Q&A, and structured response generation with fallback rules.
- Integrated the pipeline with company knowledge base and product catalogs, enabling context-aware, personalized replies and automated issue resolution.
- Applied Pydantic validation and logging to ensure message consistency, achieving a 65% reduction in support resolution time and improving client satisfaction scores.
- Deployed the chatbot as a FastAPI service on AWS EC2, connected to WhatsApp Business API, scalable messaging.
Technologies Used: Python, LangChain, GPT-4o, FastAPI, Pydantic, WhatsApp API, AWS EC2

Freelance Experience

Dec 2022 – May 2024

Legal Document Intelligence Chatbot

- Developed a Streamlit-based legal assistant using a locally hosted LLaMA model with Retrieval-Augmented Generation: indexed and chunked case documents in a FAISS vector store, implemented LangChain chains for context-aware Q&A, and logged interactions for continuous knowledge-base improvements.
Technologies Used: Python, Streamlit, LLaMA (local), LangChain, FAISS

Fitness Form Analyzer

- Engineered a real-time fitness monitoring tool for yoga instructors using OpenCV and MediaPipe: captured webcam feeds to detect and count squats and lunges, overlaid live rep counts and corrective prompts, and stored performance metrics in MySQL for progress tracking.
- Extracted human pose keypoints via MediaPipe to drive form-correction logic and ensure exercise accuracy.
Technologies Used: Python, OpenCV, MediaPipe, MySQL, Streamlit

E-commerce Image Classification Pipeline

- Built a Django-based image classification service using PyTorch transfer learning to categorize product images and integrate SKU metadata.
- Integrated MediaPipe keypoints for precise SKU-pose mapping, automating filename conventions based on detected keypoints and SKU codes.
Technologies Used: Python, Django REST Framework, PyTorch, MediaPipe, OpenCV, MySQL

EDUCATION

Bachelors in Computer Applications (BCA), Swami Vivekanand Subharti University, Meerut

CERTIFICATIONS

- **LLMOps Specialization** – Duke University (Coursera), 2025
- Data Science Post Graduate Certification – Internshala Trainings
- Python Bootcamp 2022 – Udemy
- Python for Computer Vision with OpenCV – Udemy
- PyTorch for Deep Learning – Udemy